SPEED RUN: SESSION 1

Finance Markets & Instruments (FMI)

Accounting meaning, Financial Ratios, Cost Structure, CVP, Time Value of Money (TVM), Capital Budgeting

Speed Run: Session 1

Topics	Block
1	Accounting foundations and the three statements
2	Financial Ratios: liquidity, leverage, profitability
3	Management accounting and CVP
4	Time value of money
5	Capital Budgeting

Accounting: Meaning

Accounting records, classifies and summarises business transactions in money terms. Every transaction keeps this balanced:

Assets = Liabilities + Equity

Ravi opens a kirana store	Cash	Inventory	Loan	Equity
Invests ₹5L, borrows ₹2L	7L	–	2L	5L
Buys stock for ₹3L cash	4L	3L	2L	5L
Sells ₹1L stock for ₹1.5L cash	5.5L	2L	2L	5.5L

Assets: What the Business Owns

An asset is an economic resource the business owns or controls that is expected to give future economic benefit.

Current assets

Realised, sold or consumed within the operating cycle or about 12 months: cash, bank balance, receivables, inventory, short-term investments, prepaid expenses.

Non-current assets

Benefit over a longer period: buildings, machinery, computers, vehicles, long-term investments, certain intangibles, right-of-use (lease) assets.

Kirana store: cash ₹50,000, stock ₹3L and customers owing ₹20,000 are current. The ₹1.2L freezer and shop fittings are non-current.

Liabilities & Equity: Who funds the assets

Current liabilities

Due within the operating cycle or about 12 months: accounts payable, salaries payable, interest payable, taxes payable, short-term borrowings.

Non-current liabilities

Due beyond 12 months: long-term bank loans, long-term bonds, certain lease liabilities.

A liability is an obligation to an outsider. Equity is the owners' residual claim: Equity = Assets − Liabilities, from capital invested and profits retained, reduced by losses and drawings or dividends.

Store example: supplier bill ₹80,000 and salary due ₹30,000 are current; a ₹5L five-year bank loan is non-current.

Accounting Equation

Assets = Liabilities + Equity

Proprietorship

Equity = Owner's capital + Profit − Drawings

Company

Equity = Share capital + Retained earnings + Other reserves

Seven elements: assets, liabilities, equity, income, expenses, gains and losses. Income, expenses, gains and losses change equity: income and gains raise it, expenses and losses reduce it, so Equity moves by Income − Expenses + Gains − Losses (before drawings or dividends).

The equation must balance after every transaction. Every ₹ of assets is funded by outsiders or by owners.

Users of Accounting Information

User	Uses statements to	Example
Management	Plan, control costs, decide	Retail chain picks best store for expansion
Employees	Judge job security, bonuses	Strong profits raise incentive hopes
Shareholders and investors	Measure ROI and growth	Review quarterly and annual results
Lenders	Check repayment ability	Bank reads audited statements pre-loan
Suppliers and creditors	Set credit limits	Supplier checks position before more credit
Government	Verify tax, compliance	GST and income tax filings

Internal users run the business; external users invest, lend, supply or regulate it. Types: financial, management, cost, tax, audit, forensic.

Core Assumptions

Concept	Meaning	Real-life example
Business entity	Business is separate from the owner	Amazon's books are not Bezos's wealth
Going concern	Business will continue operating	Oven valued at cost less wear, not scrap
Money measurement	Record only what is measurable in money	A skilled team is not an asset line
Accounting period	Results measured per period	Indian year: 1 April 2026 to 31 March 2027
Historical cost	Assets recorded at acquisition cost	Land bought at ₹50L stays at cost

Next: four concepts that drive profit measurement: accrual, dual aspect, matching and realisation.

Accrual concepts: Record As & When it Happens

Record income and expense when the economic activity occurs, not when cash moves.

Event	Economic month	Cash month	Treatment
ABC Ltd ad campaign ₹1,00,000	March 2026	April 2026	March expense; ₹1L payable at 31 March
Salary for March, paid 5 April	March	April	March expense; salary payable
Annual insurance ₹12,000 paid 1 April	12 months	April	₹1,000 a month; rest is prepaid

Cash-basis view would show no March ad cost and an April cost: profit for both months would be misstated.

Dual Aspect: every transaction has two effects

Transaction	Effect 1	Effect 2
Owner invests ₹5L cash	Cash +5L (asset)	Equity +5L
Buy machine ₹2L cash	Machine +2L (asset)	Cash −2L (asset)
Buy stock ₹1L on credit	Inventory +1L (asset)	Payables +1L (liability)
Pay supplier ₹60,000	Payables −60,000	Cash −60,000

This is the basis of double-entry bookkeeping. After each line, Assets = Liabilities + Equity still holds.

Matching Concept: Pair cost with the revenue it earned

FashionHub buys 1,000 shirts at ₹1,000 and sells 600 at ₹1,800.

Item	Working	Amount
Sales	600 × ₹1,800	₹10,80,000
Cost of goods sold	600 × ₹1,000	₹6,00,000
Gross profit	₹10,80,000 − ₹6,00,000	₹4,80,000
Closing inventory (asset)	400 × ₹1,000	₹4,00,000

Using the full ₹10L purchase as an expense would show a loss of ₹(0.8L) and hide ₹4L of unsold stock.

Realisation: Revenue when it is earned

A ₹12,000 Netflix plan paid on 1 April 2025 is not ₹12,000 revenue that day. Service is delivered over 12 months, so ₹1,000 is recognised each month.

Date	Revenue recognised so far	Unearned revenue (liability)
30 June 2025	₹3,000	₹9,000
30 September 2025	₹6,000	₹6,000
31 March 2026	₹12,000	₹0

Same logic: airline tickets, coaching fees and annual software licences. Recognise revenue when the performance obligation is met, not when cash arrives.

Four conventions that guide judgment

Consistency

Keep the same method period to period. Changing depreciation yearly to shape profit destroys comparability.

Prudence

Be cautious under uncertainty: do not overstate assets or income.

Full disclosure

Give users the material information needed to understand position and performance.

Materiality

Information matters if omitting or misstating it could influence users' decisions.

Transactions and the equation: GreenBite Cafe

Transaction	A	L	E	Total assets	Equity
1. Ananya invests ₹8L	+	–	+	₹8.00L	₹8.00L
2. Bank loan ₹4L	+	+	–	₹12.00L	₹8.00L
3. Catering ₹1.2L on credit, ₹50,000 expenses	+	–	+	₹12.70L	₹8.70L
4. Sell fridge ₹90,000, carrying ₹70,000 (gain ₹20,000)	+	–	+	₹12.90L	₹8.90L
5. Rent ₹40,000; sell machine ₹30,000, carrying ₹45,000 (loss ₹15,000)	+	–	+	₹12.35L	₹8.35L

Loan ₹4L stays constant, so Assets ₹12.35L = Liabilities ₹4L + Equity ₹8.35L. Income, expenses, gains and losses all flow through equity.

Example: Rahul's Cafe (a proprietorship)

Step	Cash (A)	Liabilities	Equity
Starts with ₹5,00,000	₹5,00,000	0	₹5,00,000
Cash expenses ₹3,00,000	₹2,00,000	0	₹2,00,000
Cash sales ₹4,00,000	₹6,00,000	0	₹6,00,000
Drawings ₹50,000	₹5,50,000	0	₹5,50,000

Cash taken for personal use is drawings: it reduces equity directly, not an expense of the business.

Example: Tech Star Ltd (Corporate)

Step	Assets	Liabilities	Equity
Issues shares for ₹10L	Cash 10L	0	10L
Credit sales ₹8L	Cash 10L + receivables 8L	0	18L
Credit expenses ₹5L	Cash 10L + receivables 8L	Payables 5L	13L
Dividend ₹1L paid	Cash 9L + receivables 8L	Payables 5L	12L

Final check: assets ₹17L = liabilities ₹5L + equity ₹12L. Credit transactions move receivables and payables, not cash.

Three Important Financial Statements

Income Statement

Performance over a period. Revenue − expenses = profit.

Balance Sheet

Position on one date. Assets = Liabilities + Equity.

Cash Flow

Cash in and out: operating, investing, financing.

How they link: profit flows into equity on the balance sheet. Depreciation lowers profit but is not a cash outflow.

Depreciation example: machine ₹10L, residual ₹1L, life 5 years → (10 − 1) ÷ 5 = ₹1.8L expense each year (straight line).

Depreciation: Spreading an asset's cost over its life

Depreciation allocates the cost of a non-current asset over the years it helps earn revenue (matching concept). Four terms:

Cost

What the machine cost: ₹11,00,000

Residual value

Expected disposal value: ₹1,00,000

Useful life

Years of use: 5

Depreciable amount

Cost − residual = ₹10,00,000

Two methods follow: straight line (same charge every year) and reducing balance (a fixed percentage of the opening book value). Example for a laptop: cost ₹60,000, residual ₹10,000, 5 years → ₹10,000 a year on straight line.

Straight-Line Method (SLM)

Annual depreciation = (Cost − Residual value) ÷ Useful life. Machine: ₹10,00,000 cost, ₹1,00,000 residual, 5 years → ₹1,80,000 every year.

Year	Opening balance	Depreciation	Closing balance
1	₹10,00,000	₹1,80,000	₹8,20,000
2	₹8,20,000	₹1,80,000	₹6,40,000
3	₹6,40,000	₹1,80,000	₹4,60,000
4	₹4,60,000	₹1,80,000	₹2,80,000
5	₹2,80,000	₹1,80,000	₹1,00,000

Reducing-Balance Method

Annual depreciation = Opening balance × rate. Truck: ₹20,00,000, 25%, residual nil.

Year	Opening balance	Depreciation at 25%	Closing balance
1	₹20,00,000	₹5,00,000	₹15,00,000
2	₹15,00,000	₹3,75,000	₹11,25,000
3	₹11,25,000	₹2,81,250	₹8,43,750
4	₹8,43,750	₹2,10,938	₹6,32,812
5	₹6,32,812	₹1,58,203	₹4,74,609

Charge is heaviest early: Year 1 is ₹5L, where straight line on the same truck would be ₹4L every year. Suits assets that lose value fast, such as vehicles.

Intangible Assets

Identifiable non-monetary assets with no physical substance that the business controls and expects to give future benefit.

Examples

Patents, copyrights, trademarks, brands, software licences, franchises, customer relationships, technology, qualifying R&D.

Recognise only if

Future benefits are probable and cost can be measured reliably. Externally bought: at cost.

Illustration: patent bought for ₹10 crore, 10-year life	Amount
Amortisation each year (straight line)	₹1 crore
Carrying amount after 3 years (cost model)	₹7 crore

Goodwill is treated separately and arises only in mergers and acquisitions.

Long-Term Investments

Company	Strategic	Financial
Tata Motors (FY25, standalone)	₹29,718 cr in subsidiaries, JVs, associates	₹1,709 cr other investments
Volkswagen (Dec 2025)	€10.0 bn equity-accounted (e.g. SAIC-VW 50%, QuantumScape 16%)	€11.7 bn other financial assets

Strategic investments support the business model; financial investments earn dividends, interest or capital gains. Tata Motors' total long-term investments: ₹31,427 crore.

Income statement: Important Line Items

Line	Meaning	Formula or example
Revenue	Income from selling goods or services, shown first	Sales ₹500L
Cost of goods sold	Direct cost of what was sold: materials, direct labour, production costs	₹300L
Gross profit	Profit after direct costs	Revenue − COGS = ₹200L
Operating expenses	Costs of running the business	Selling and general ₹80L + depreciation ₹20L
Operating profit	Profit from the main business	Gross profit − operating expenses = ₹100L

Selling expenses: commission, advertising, distribution. General expenses: management salaries, rent, insurance, audit, legal, utilities. Figures are for an illustrative company, in ₹ lakh.

Income statement: Important Line Items

Line	Meaning	Formula or example
Other income	Non-operating income: interest, dividends, profit on asset sales	Interest income ₹10L
Finance costs and other expenses	Interest on loans, finance charges, losses on asset sales	Interest ₹20L
Profit before tax	Operating profit + other income − other expenses	100 + 10 − 20 = ₹90L
Income tax	Tax on taxable income	25% of ₹90L = ₹22.5L
Net income (PAT)	PBT − tax	₹67.5L
Earnings per share	Net profit ÷ shares outstanding	₹67.5L ÷ 10 lakh shares = ₹6.75

Example: Income Statement (₹ lakh)

Item	₹ lakh	Margin
Revenue	500.0	100%
Cost of goods sold	(300.0)	
Gross profit	200.0	40%
Selling and general expenses	(80.0)	
Depreciation	(20.0)	
Operating profit	100.0	20%
Other income	10.0	
Interest (finance cost)	(20.0)	
Profit before tax	90.0	
Income tax at 25%	(22.5)	
Net income	67.5	13.5%
EPS (10 lakh shares)	₹6.75	

Cash flow Statement: Where cash came from and went

It shows inflows and outflows of cash and cash equivalents in a period. The income statement shows profit; this statement explains the change in cash.

Operating

Day-to-day business. In: cash from customers and services. Out: suppliers, salaries, rent and other expenses, taxes.

Investing

Long-term assets. Out: buying machinery, buildings, long-term investments. In: selling them.

Financing

Capital. In: shares, debentures, bank loans. Out: buybacks, dividends, interest, loan repayment.

Why Net Income & Cash Differ

Event	Effect on net income	Effect on cash
Credit sale not yet collected	Profit up	No cash yet
Depreciation	Profit down	No cash out
Inventory bought, not yet sold	No effect yet	Cash out
Expenses accrued, supplier unpaid	Profit down	No cash out yet
Machine purchase	Spread as depreciation	Full cash out now
Loan repayment or dividend	Not an expense	Cash out

Accrual accounting records activity; the cash statement records money. A profitable business can still run short of cash.

Reconciling Net Income to Operating Cash flow

Start: net income	₹67.5L
Add: depreciation (non-cash)	+ ₹20.0L
Less: increase in receivables	− ₹30.0L
Less: increase in inventory	− ₹15.0L
Add: increase in payables	+ ₹10.0L
Cash from operating activities	₹52.5L

Rules: add back non-cash expenses (depreciation, amortisation, stock-based pay, deferred tax). Increases in receivables or inventory use cash, so subtract. Increases in payables keep cash in, so add. Decreases do the reverse.

Putting the three activities together (₹ lakh)

Activity	Item	₹ lakh
Operating	From net income reconciliation	52.5
Investing	Purchase of machinery	(40.0)
Financing	Bank loan 20.0, dividend paid (5.0)	15.0
Net change in cash	52.5 − 40.0 + 15.0	27.5
Closing cash	Opening 10.0 + 27.5	37.5

Net income was ₹67.5L, yet cash rose by only ₹27.5L: receivables, inventory and machinery absorbed cash.

Financial Ratios: Current Ratio

Current ratio

Current assets ÷ current liabilities. Rule of thumb 1.5–2. Example: ₹60L ÷ ₹40L = 1.5.

Quick ratio

(Current assets − inventory) ÷ current liabilities. (60 − 20) ÷ 40 = 1.0.

Cash ratio

Cash and equivalents ÷ current liabilities. ₹8L ÷ ₹40L = 0.2.

Debt-to-equity

Total debt ÷ equity. ₹50L ÷ ₹100L = 0.5. Higher means more risk.

Interest coverage

EBIT ÷ interest: ₹24L ÷ ₹6L = 4×. Below 2 is a warning sign.

Five families: liquidity, efficiency, leverage, profitability, market.

Efficiency Ratios: How fast does money cycle?

Inventory days

Days stock sits before sale. COGS ₹140L ÷ stock ₹20L = 7×, so 365 ÷ 7 = 52 days.

Receivable days

Days customers take to pay. Sales ₹200L ÷ debtors ₹25L = 8×, so 46 days.

Payable days

Days we take to pay suppliers. COGS ₹140L ÷ creditors ₹28L = 5×, so 73 days.

Operating cycle = Inventory days + Receivable days − Payable days

Example: 45 + 30 − 35 = 40 days of cash tied up. Shorten it by collecting faster or stretching suppliers within terms. Solvency ratios (D/E, interest cover) cover long-term obligations.

Profitability Ratios: How well do we earn?

Gross margin

(Sales − COGS) ÷ Sales: (200 − 140) ÷ 200 = 30%

Operating margin

EBIT ÷ Sales: 24 ÷ 200 = 12%

Net margin

PAT ÷ Sales: 14 ÷ 200 = 7%

ROCE

EBIT ÷ capital employed: 24 ÷ 150 = 16%

ROE

PAT ÷ equity: 14 ÷ 100 = 14%

Asset turnover

Sales ÷ total assets: 200 ÷ 190 = 1.05×

DuPont: ROE = Net margin × Asset turnover × Equity multiplier. Retail wins on turnover, luxury on margin.

Market Ratios: What do investors pay for?

Measure	Formula	Tells you
EPS	Profit to ordinary shareholders ÷ shares	Profit per share
P/E	Market price ÷ EPS	Price paid per ₹1 of earnings
DPS, payout	DPS = dividend ÷ shares; payout = dividend ÷ profit	Cash returned
Dividend yield	DPS ÷ market price	Income return on price
Market cap	Price × shares outstanding	Equity value
Enterprise value	Market cap + debt − cash	Whole-business value (M&A)

Example: PAT ₹14L, 1L shares → EPS ₹14. Price ₹210 → P/E 15×. DPS ₹4 → payout 29%, yield 1.9%.

Example: Asian Paints Ltd (Standalone, FY2025)

Income statement (₹ crore)	FY2025	Balance sheet (₹ crore)	31 Mar 2025
Net sales	29,552.65	Inventories	5,848.23
Cost of sales (proxy)	17,340.54	Sundry debtors	3,197.55
Other income	770.31	Cash and bank	204.50
Depreciation	901.85	Total current assets	11,175.61
Interest	143.77	Current liabilities and provisions	5,973.52
Profit before tax	4,901.60	Debt	1,198.82
Net profit	3,588.07	Shareholders' funds	18,987.75
EPS (₹) / DPS (₹)	37.41 / 24.80	Total assets	20,464.73

Source: market-data aggregators (CMOTS-based) for the year ended 31 March 2025; check against the annual report. Cost of sales = raw materials + power and fuel + other manufacturing expenses − stock adjustment. EBIT = PBT + interest = ₹5,045.37 cr, which includes ₹770.31 cr other income.

Liquidity Ratios: Example

Ratio	Formula	Working (₹ cr)	Result
Current ratio	Current assets ÷ current liabilities	11,175.61 ÷ 5,973.52	1.87×
Quick ratio	(Current assets − inventory) ÷ current liabilities	(11,175.61 − 5,848.23) ÷ 5,973.52	0.89×
Cash ratio	Cash and equivalents ÷ current liabilities	204.50 ÷ 5,973.52	0.03×

Reading it: each ₹1 of short-term obligations is backed by ₹1.87 of current assets. Stripping out inventory, cover falls to ₹0.89, and cash alone covers only about 3 paise. The current ratio was 1.92× a year earlier, so liquidity is stable. Low cash is not alarming for a firm that collects from dealers and keeps strong investments elsewhere.

Efficiency Ratio: Example

Ratio	Formula	Working (₹ cr)	Result
Inventory turnover	Cost of sales ÷ average inventory	17,340.54 ÷ 5,501.91	3.15×
Inventory days	365 ÷ turnover	365 ÷ 3.15	116 days
Receivables turnover	Net sales ÷ average debtors	29,552.65 ÷ 3,448.70	8.57×
Collection days	365 ÷ turnover	365 ÷ 8.57	43 days

Reading it: stock sits for about 116 days before it is sold and customers pay in about 43 days. Fewer inventory days or collection days free cash. Debtors fell from ₹3,699.84 cr to ₹3,197.55 cr while sales dipped, which helped collection days.

Efficiency Ratio: Example

Ratio	Formula	Working (₹ cr)	Result
Payables turnover	Cost of sales ÷ average creditors	17,340.54 ÷ 3,387.19	5.12×
Payable days	365 ÷ turnover	365 ÷ 5.12	71 days
Operating cycle	Inventory days + receivable days − payable days	116 + 43 − 71	87 days

Reading it: the business waits about 87 days between paying suppliers and collecting from customers, funded by working capital. Longer supplier credit or faster stock turns shorten the cycle; a shorter cycle means less cash tied up.

Solvency Ratios: Example

Ratio	Formula	Working (₹ cr)	Result
Debt-to-equity	Interest-bearing debt ÷ equity	1,198.82 ÷ 18,987.75	0.06×
Interest coverage	EBIT ÷ interest expense	5,045.37 ÷ 143.77	35.1×

Reading it: debt is about 6 paise per ₹1 of equity (it was 0.07× a year earlier), and operating profit covers interest about 35 times. This is a lightly geared balance sheet, so financial risk is low. Contrast a company with D/E above 2 and coverage under 2.

Profitability Ratio: Example

Ratio	Formula	Working (₹ cr)	FY2025
Gross profit margin	(Sales − cost of sales) ÷ sales	(29,552.65 − 17,340.54) ÷ 29,552.65	41.3%
Operating profit margin	EBIT ÷ sales	5,045.37 ÷ 29,552.65	17.1%
Net profit margin	Net profit ÷ sales	3,588.07 ÷ 29,552.65	12.1%

Reading it: of every ₹100 of sales, about ₹41 remains after direct costs and ₹12 reaches shareholders. Compare FY2024: gross margin 42.4% and net margin 17.0%. Margins fell because costs rose while sales slipped. Operating margin is 17.1% including other income; excluding it, about 14.5%.

Return Ratios: Example

Ratio	Formula	Working (₹ cr)	Result
ROCE	EBIT ÷ (equity + debt)	5,045.37 ÷ 20,186.57	25.0%
ROA	Net profit ÷ total assets	3,588.07 ÷ 20,464.73	17.5%
ROE	Net profit ÷ shareholders' equity	3,588.07 ÷ 18,987.75	18.9%

Reading it: each ₹100 of long-term capital earns about ₹25 before interest and tax, and each ₹100 of shareholders' money earns about ₹19. ROE is close to ROA because there is little debt, so leverage is not inflating returns. ROI is a broader idea: gain ÷ cost, for example ₹1,00,000 growing to ₹1,25,000 is 25%.

Market Ratios: Example

Ratio	Formula	Working	Result
EPS	Net profit ÷ shares	₹3,588.07 cr ÷ 95.92 cr shares	₹37.41
P/E	Price ÷ EPS	₹2,400 ÷ ₹37.41	64.2×
Dividend per share	Dividend ÷ shares	as declared	₹24.80
Payout ratio	DPS ÷ EPS	24.80 ÷ 37.41	66%
Dividend yield	DPS ÷ price	24.80 ÷ 2,400	1.03%

Reading it: at an assumed price of ₹2,400, investors pay about ₹64 for each ₹1 of earnings, a high multiple that prices in growth. Two-thirds of earnings is paid out, giving a cash yield near 1%.

Market Value: Capitalisation & Enterprise Value

Measure	Formula	Working	Result
Market capitalisation	Price × shares	₹2,400 × 95.92 cr	₹230,208 cr
Enterprise value	Market cap + debt − cash	230,208 + 1,198.82 − 204.50	₹231,202 cr
Price-to-book	Price ÷ book value per share	2,400 ÷ 197.95	12.1×

Reading it: the market values the whole business near ₹2.3 lakh crore; EV is only slightly above market cap because debt and cash are small. A price-to-book of 12× shows investors value brand and earning power far above accounting book value.

Summary: Financial Ratios

Family	Key ratio	Asian Paints FY2025	Message
Liquidity	Current ratio	1.87×	Comfortable cover
Efficiency	Operating cycle	87 days	Stock-heavy, suppliers fund part
Solvency	D/E; interest cover	0.06×; 35×	Very low financial risk
Profitability	Gross; net margin	41%; 12%	Strong, but slipped from FY2024
Returns	ROCE; ROE	25%; 19%	High returns without debt
Market	P/E; yield	64×; 1.0%	Premium valuation

Best practice: compare over time, with peers and with budget, and never judge a company on one ratio.

Management Accounting: From Information to Decisions

Basis	Financial accounting	Management accounting
Users	Investors, lenders, regulators	Managers inside the firm
Rules	GAAP and accounting standards	No fixed rules; built for the decision
Time view	Past results	Future-oriented plans and options
Detail	Whole company	Products, departments, projects, customers

Planning

Estimate costs, revenues, profit and resources: budgets.

Control

Compare actual with budget and study variances. Budget ₹50L, actual ₹54L → ₹4L adverse variance.

Decisions

Make or buy, outsource or produce, product choice, and pricing (cost-plus, penetration, skimming, marginal-cost).

Cost Concepts

Term	Meaning	Example
Cost	Sacrifice of resources to achieve an objective	₹10L paid for a machine
Expense	Cost charged against revenue in a period	₹1.8L yearly depreciation
Actual vs budgeted	Incurred cost vs estimated cost	Actual ₹54L vs budget ₹50L
Cost object	Anything we measure cost for	Product, project, customer, department
Cost unit	Unit in which cost is expressed	Per car; per room-night; per kWh; per patient-day

Unit cost: total production cost ₹50L ÷ 10,000 units = ₹500 a unit. Airlines use cost per passenger-kilometre; transport firms use cost per tonne-kilometre.

Management Accounting: Costs Classification

Financial accounting reports to outsiders under GAAP. Management accounting serves managers: plan, control, decide (make or buy, pricing, product mix).

Classify by	Categories	Car factory example
Nature	Material, labour, other	Steel, assembly wages, rent
Behaviour	Fixed, variable, semi-variable	Rent vs steel vs electricity bill
Traceability	Direct, indirect	Steel body vs factory lubricant
Function	Production, selling, admin	Plant cost vs showroom cost

Unit cost: total production cost ₹50L ÷ 10,000 units = ₹500 per unit.

Classifying Costs By Nature & Traceability

Basis	Category	Real-life example (car factory)
Nature	Direct material	Steel body of one car
Nature	Indirect material	Lubricants used in the factory
Nature	Direct labour	Assembly-line worker
Nature	Indirect labour	Factory security guard, supervisor
Traceability	Direct cost	Steel or wages traced to a specific car
Traceability	Indirect cost (overhead)	Factory rent, electricity, insurance

Classifying Costs: By Function

Function	What it covers	Examples
Production	Making the product	Raw materials, factory wages, factory rent, electricity, depreciation
Administration	Running the organisation	Senior management salary, audit fees, legal fees, head-office rent
Selling	Winning sales	Sales commission, advertising, promotion
Distribution	Getting goods to customers	Warehousing, transport, delivery charges

Only production costs go into product (inventory) cost under absorption costing; administration, selling and distribution are period costs.

Cost Classification: By Behaviour

Type	Example	What happens as volume rises
Fixed	Factory rent ₹10L a month	Total flat; per car ₹1,000 at 1,000 cars, ₹100 at 10,000
Variable	Raw material ₹20 per bottle	Total ₹20,000 at 1,000 bottles, ₹2,00,000 at 10,000; per unit stays ₹20
Semi-variable	Power: ₹50,000 fixed + ₹8 per machine-hour	₹58 per hour at 1,000 hours, ₹13 at 10,000

Relevance: only future costs that differ between options matter. A ₹2 crore software build already spent is a sunk cost: ignore it, and compare only the cost and revenue of launching versus abandoning.

Classifying Costs By Relevance: Only What matters to a decision

Relevant cost

Future cost that differs between options.

Irrelevant cost

Cost that is the same whichever option is chosen.

Sunk cost

Already spent and unrecoverable. ₹2 crore spent building software is ignored when choosing to launch or abandon it.

Opportunity cost

Benefit of the best alternative forgone. Choosing an MBA over a ₹16 lakh job costs ₹16 lakh of salary.

Absorbing Overheads into Product Cost

Total production cost = Direct material + Direct labour + Production overheads. Unit cost is needed to value stock, set prices, measure profit, compare products and decide to make or outsource.

The question

A motor uses ₹4,000 direct material and ₹2,000 direct labour (₹6,000 direct). Factory rent, power, depreciation, supervisor pay, maintenance and security also exist. How much belongs to each motor?

Absorption

Spread overheads over units using an absorption rate: OAR = Total overhead ÷ activity base (units, labour hours or machine hours).

Source example: overhead ₹1,00,000 ÷ 20,000 units = ₹5 absorbed into every unit.

Overhead Absorption: Example

Step	Working	Result
Overhead absorption rate	₹10,00,000 overhead ÷ 10,000 motors	₹100 a motor
Full product cost	₹4,000 + ₹2,000 + ₹100	₹6,100
Profit at ₹7,500 price	₹7,500 − ₹6,100	₹1,400 a motor
If actual overhead is ₹10,50,000	₹10,50,000 − ₹10,00,000 absorbed	₹50,000 under-absorbed

Under-absorbed overhead (actual above absorbed) reduces profit; over-absorbed overhead increases it. It is adjusted in the income statement.

Marginal Cost & Marginal Costing

Marginal cost is the extra cost of producing one more unit; in practice, the variable cost per unit while fixed costs stay unchanged.

Illustration	Working	Result
Output rises 10,000 → 10,100	+100 units	+100 units
Total cost rises ₹50,00,000 → ₹50,60,000	+₹60,000	+₹60,000
Marginal cost per component	₹60,000 ÷ 100	₹600

Marginal costing charges only variable costs to products; fixed costs are period costs. Its central measure is contribution = Selling price − marginal cost. A ₹900 component earns ₹300 contribution (added illustration).

Absorption vs Marginal Costing:

Production 10,000, sales 8,000 at ₹500	Absorption	Marginal
Sales	₹40,00,000	₹40,00,000
Cost of goods sold / variable cost (₹300 + ₹100 fixed vs ₹300)	(₹32,00,000)	(₹24,00,000)
Gross profit / contribution	₹8,00,000	₹16,00,000
Fixed production overhead charged in full	–	(₹10,00,000)
Selling and admin overhead	(₹2,00,000)	(₹2,00,000)
Profit	₹6,00,000	₹4,00,000

The ₹2,00,000 gap is ₹100 of fixed overhead on each of the 2,000 unsold units, carried in closing stock under absorption costing.

Contribution:

An empty seat normally sells at ₹8,000. Aircraft lease, crew and airport charges are already committed.

Extra cost of carrying one passenger	₹
Meal	500
Passenger service	200
Processing	100
Other variable cost	200
Marginal cost	1,000

A last-minute buyer pays ₹3,000: contribution = ₹3,000 − ₹1,000 = ₹2,000. With no capacity constraint, sell the seat.

Contribution with a capacity constraint

Flight of 180 seats, 175 sold at ₹8,000, so 5 seats remain. A travel firm wants 10 seats at ₹3,000; individuals would pay ₹7,000.

Question	Working	Answer
Contribution, travel firm	₹3,000 − ₹1,000	₹2,000 a seat; ₹10,000 for 5
Contribution, individual	₹7,000 − ₹1,000	₹6,000 a seat; ₹30,000 for 5
Opportunity cost of a seat	Contribution forgone	₹6,000
Minimum acceptable price	₹1,000 + ₹6,000	₹7,000

Decision: reject ₹3,000. Accepting loses ₹20,000 of contribution: 5 × (₹6,000 − ₹2,000).

Cost Volume Profit (CVP) Analysis

Questions

How many units to break even? To earn a target profit? What if volume, price, variable cost or fixed cost changes? How far can sales fall before a loss?

Key terms

Selling price (SP), variable cost (VC), contribution = SP − VC, fixed cost (FC), break-even point (BEP).

At break-even, total revenue = total cost, so profit is zero. Contribution covers fixed costs first, then adds to profit. Assumes costs split cleanly into fixed and variable, and constant price and unit variable cost within the relevant range.

CVP Analysis

Contribution

Price − variable cost per unit: ₹500 − ₹300 = ₹200

P/V ratio

Contribution ÷ Sales: 200 ÷ 500 = 40%

Break-even

Fixed ₹10L ÷ ₹200 = 5,000 units

Target profit units

(₹10L + ₹4L target) ÷ ₹200 = 7,000 units

Margin of safety

8,000 − 5,000 = 3,000 units, 37.5% of sales

Marginal costing: Sales − variable cost = contribution; contribution − fixed cost = profit.

CVP Formula

Question	Working	Answer
Contribution	₹500 − ₹300 per unit × 10,000 units	₹20L
Break-even in sales value	Fixed ₹2L ÷ P/V 40%	₹5L
Margin of safety (units)	10,000 − 6,000	4,000 units
Margin of safety (value)	₹10L − ₹6L actual vs break-even	₹4L, 40% of sales
Units for ₹8L profit	(₹12L + ₹8L) ÷ ₹200	10,000 units

Chart check: price $50, variable cost $30, fixed $40,000 → break-even 2,000 units ($100,000). At 3,000 units profit is $20,000 and margin of safety 1,000 units.

Margin of Safety: How Far Can Sales Fall

Measure	Formula	Source example
MoS in units	Actual units − break-even units	10,000 − 6,000 = 4,000 units
MoS in value	Actual sales − break-even sales	₹10,00,000 − ₹6,00,000 = ₹4,00,000
MoS %	MoS ÷ actual sales	₹4,00,000 ÷ ₹10,00,000 = 40%
Shortcut	Profit ÷ P/V ratio	Gives MoS in value

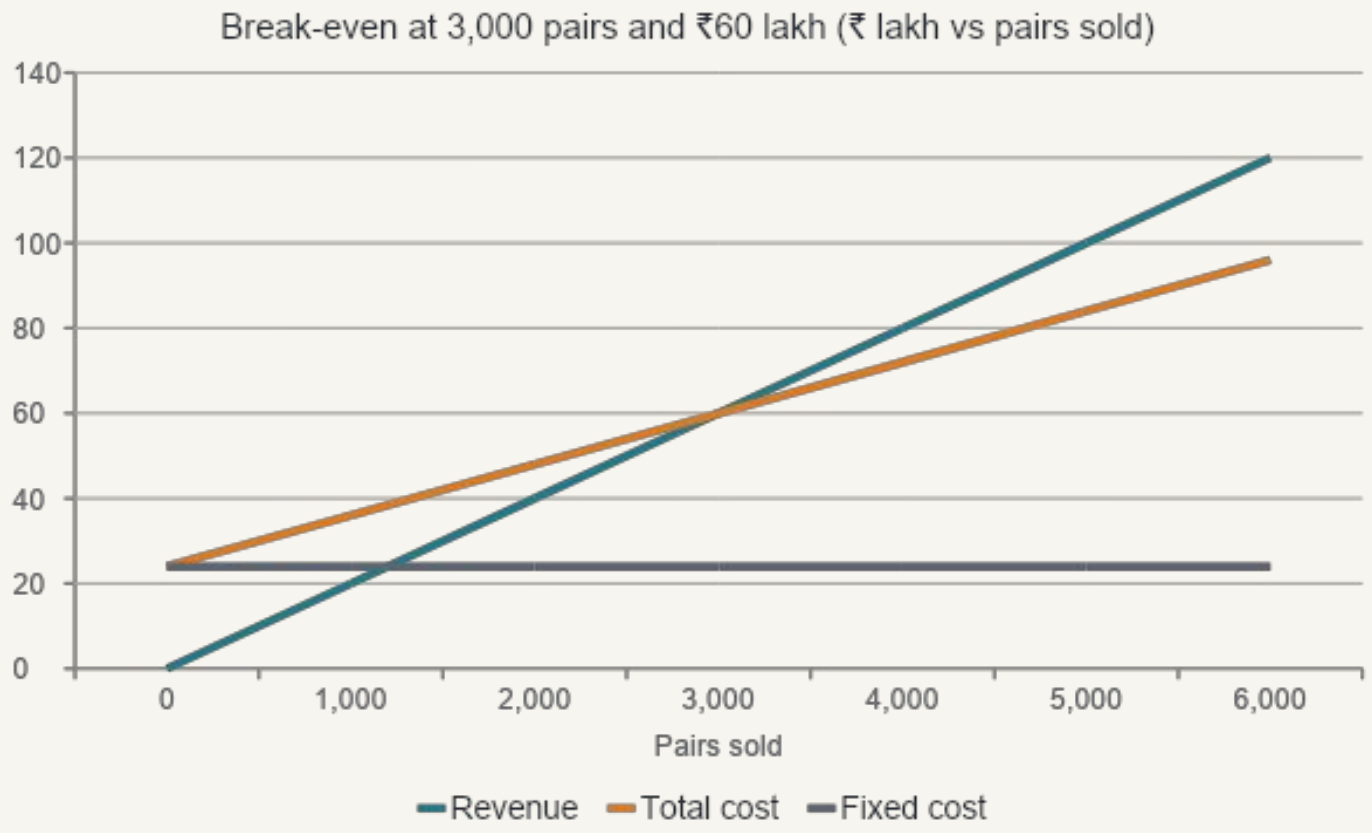
A higher margin of safety means lower risk of falling below break-even. Here sales can fall 40% before the business starts losing money.

Example: ClassicShoes Pvt Ltd

Price ₹2,000, variable cost ₹1,200 (material 700, labour 300, overhead 200), fixed cost ₹24,00,000 a year.

Measure	Working	Result
Contribution per pair	₹2,000 − ₹1,200	₹800
P/V ratio	₹800 ÷ ₹2,000	40%
Break-even (units)	₹24,00,000 ÷ ₹800	3,000 pairs
Break-even (value)	₹24,00,000 ÷ 40%	₹60,00,000
Profit at 4,500 pairs	4,500 × ₹800 − ₹24,00,000	₹12,00,000
Margin of safety	1,500 pairs; ₹30,00,000	33.3% of sales

Break-even chart for ClassicShoes



₹12L
Profit at 4,500 pairs

1,500 pairs
Margin of safety (33%)

Loss
below 3,000 pairs

ClassicShoes: What If?

Scenario	New contribution	Break-even pairs
Base case	₹800	3,000
Price cut to ₹1,800	₹600	4,000
Variable cost up 10% (₹1,320)	₹680	3,529
Fixed cost up ₹3,00,000	₹800	3,375
Sales fall 20% to 3,600 pairs	₹800	profit drops to ₹4,80,000

A price cut hurts most: break-even rises by a third. Ask whether volume gains would offset it.

Example: A cloud kitchen

Price ₹250 a meal, variable cost ₹160 (ingredients, packaging, delivery commission), fixed cost ₹6,00,000 a month.

Measure	Working	Result
Contribution	₹250 − ₹160	₹90 a meal
P/V ratio	₹90 ÷ ₹250	36%
Break-even (meals)	₹6,00,000 ÷ ₹90	6,667 meals
Break-even (value)	₹6,00,000 ÷ 36%	₹16.67 lakh
Profit at 9,000 meals	9,000 × ₹90 − ₹6,00,000	₹2,10,000
Margin of safety	2,333 meals; ₹5.83 lakh	25.9%
Meals for ₹4,50,000 profit	(₹6,00,000 + ₹4,50,000) ÷ ₹90	11,667 meals

Finance: Why Money has Time Value

₹10,000 today or ₹11,000 in a year? At 10% interest, ₹10,000 grows to ₹11,000: indifferent. If the phone you want rises 20% to ₹12,000, taking ₹10,000 today and investing still leaves you short, so inflation matters.

Year	Simple interest value	Compound value
1	₹110.00	₹110.00
2	₹120.00	₹121.00
3	₹130.00	₹133.10
5	₹150.00	₹161.05

₹100 at 10% earns ₹50 simple but ₹61.05 compound in 5 years. Four drivers: preference for present consumption, investing, inflation, opportunity cost.

Time Value of Money (TVM)

Symbol	Meaning
PV / FV	Value today / value at a future date
CF	Cash flow at a point in time
r	Interest, discount or required rate per period
n	Number of periods
A	Equal periodic payment (annuity)
g	Constant growth rate of cash flows

Compounding

PV → FV: $FV = PV \times (1 + r)^n$. Factor $(1 + r)^n$.

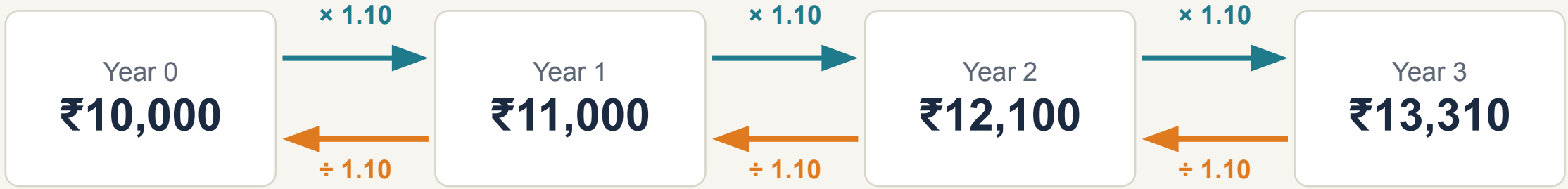
Discounting

FV → PV: $PV = FV \div (1 + r)^n$. Factor $1 \div (1 + r)^n$.

Compounding & Discounting

Compounding: PV to FV (forward)

Discounting: FV to PV (backward)



₹10,000 at 10% grows by ₹1,000, ₹1,100 and ₹1,210 in successive years: interest earns interest. Reading right to left, ₹13,310 in year 3 is worth ₹10,000 today.

Time Value of Money: A Rupee today is more Valuable than A Rupee Later

Idea	Formula	Use
Future value	$FV = PV \times (1 + r)^n$	Compounding forward
Present value	$PV = FV \div (1 + r)^n$	Discounting back
Annuity PV	$PMT \times [1 - (1 + r)^{-n}] \div r$	Loans, EMIs, leases
Annuity FV	$PMT \times [(1 + r)^n - 1] \div r$	SIPs, savings
Perpetuity	$PMT \div r$	Preference shares, endowments

Rule of 72: years to double ≈ 72 ÷ interest rate %. At 8% money doubles in about 9 years.

Example : Patel Industries' Replacement Fund

Patel Industries will need ₹50,00,000 in 5 years to replace a plant. It can earn 8% a year.

Question	Working	Answer
Compounding: ₹10,00,000 invested today grows to	₹10,00,000 × 1.08^5	₹14,69,328
Discounting: lump sum needed today for ₹50L	₹50,00,000 ÷ 1.08^5	₹34,02,916
Or equal yearly deposits (annuity)	₹50,00,000 × 0.08 ÷ (1.08^5 − 1)	₹8,52,282 a year

Interpretation: the firm can fund the plant with ₹34,02,916 now or about ₹8,52,282 at each year-end. Compounding and discounting are the same relationship read in opposite directions.

Example: GreenCart's investor offer

An investor offers GreenCart ₹8,00,000 today, or ₹10,50,000 in 3 years. GreenCart can earn 12% a year.

Method	Working	Result
Discount the future offer	₹10,50,000 ÷ 1.12^3	₹7,47,369 today
Or compound today's offer	₹8,00,000 × 1.12^3	₹11,23,942 in 3 years

Decision: ₹8,00,000 today beats ₹10,50,000 in 3 years, because the later sum is worth only ₹7,47,369 now. Both methods agree. A higher required return would make waiting even less attractive.

Single Cash Flow: Present Value

Example	Working	Present value
1. ₹50,000 due in 5 years at 8%	₹50,000 ÷ 1.08^5	₹34,029
2. ₹9,000 due in 5 years at 16.5%	₹9,000 ÷ 1.165^5	₹4,193.85

Interpretation: ₹34,029 today is equivalent to ₹50,000 in 5 years. The higher the rate, the lower the present value: at 16.5% the ₹9,000 is worth less than half its face value.

Forward view: ₹10,000 at 10% for 3 years grows to ₹13,310. A credit card at 2% a month turns ₹100 into ₹126.82, an APR of 26.82%.

Uneven Cash Flows

Discount each cash flow by its own year, then add: PV = Σ CF□ ÷ (1 + r)^{t}.

Example 1: machine at 10%	Cash flow	PV
Year 1	₹5,000	₹4,545
Year 2	₹6,000	₹4,959
Year 3	₹7,000	₹5,260
Year 4	₹8,000	₹5,464
Total		₹20,228

Example 2: a new cafe outlet at 12% returns ₹3,00,000, ₹4,00,000 and ₹5,00,000 in years 1 to 3: ₹2,67,857 + ₹3,18,878 + ₹3,55,890 = ₹9,42,625. Pay less than that and the outlet adds value.

Annuity: Equal Cash Flows for a Fixed Period

Example	Working	Result
1. ₹10,000 a year for 5 years at 10% (ordinary): PV	10,000 × 3.7908	₹37,908
1. Same deposits: future value	10,000 × 6.1051	₹61,051
2. Lease ₹2,00,000 a year for 4 years at 9%, paid at year-end: PV	₹2,00,000 × 3.2397	₹6,47,944
2. Same lease paid at the start (annuity due)	₹6,47,944 × 1.09	₹7,06,259

Ordinary annuity: payments at period end (EMIs, bond coupons). Annuity due: at period start (rent, insurance), worth (1 + r) times more. PV = A × [1 − (1 + r)$^{-n}$] ÷ r; FV = A × [(1 + r)^{n} − 1] ÷ r.

Perpetuity: Cash flows Forever

Type	Example	Working	PV
Level, PV = CF ÷ r	1. Preference share pays ₹500 a year, r = 10%	500 ÷ 0.10	₹5,000
Level	2. Endowment pays ₹6,00,000 a year, r = 8%	6,00,000 ÷ 0.08	₹75,00,000
Growing, PV = CF_1 ÷ (r − g)	1. First payment ₹500, g = 4%, r = 10%	500 ÷ 0.06	₹8,333
Growing	2. First payment ₹6,00,000, g = 5%, r = 10%	6,00,000 ÷ 0.05	₹1,20,00,000

Growing perpetuity needs r > g. Growth adds a lot of value: the same ₹6,00,000 is worth ₹75 lakh level but ₹1.2 crore growing at 5%.

Terminal Value (TV): Value beyond the forecast period

$TV_n = CF_n \times (1 + g) \div (r - g)$, valued at year n. Then discount it: $PV = TV_n \div (1 + r)^n$.

Example	Step 1: terminal value	Step 2: present value
1. Year-5 cash flow ₹2,00,000, g = 3%, r = 12%	2,06,000 ÷ 0.09 = ₹22,88,889	22,88,889 ÷ 1.12^5 = ₹12,98,590
2. Year-5 cash flow ₹50,00,000, g = 4%, r = 11%	52,00,000 ÷ 0.07 = ₹7,42,85,714	÷ 1.11^5 = ₹4,40,84,956

Interpretation: terminal value is a large share of a project's total value, so small changes in g or r move the answer a lot. Always discount it back from year n.

Worked example: four questions at 8%

₹1,46,933
₹1L invested for 5 years (FV)

₹68,058
Value today of ₹1L due in 5 years (PV)

₹39,927
PV of ₹10,000 a year for 5 years

₹58,666
FV of a ₹10,000 yearly SIP after 5 years

Perpetuity: ₹10,000 forever at 8% is worth ₹1,25,000 today. Which would you rather have: ₹1L now or ₹1.2L in 3 years? PV of ₹1.2L is ₹95,260, so take the ₹1L.

Capital Budgeting: Techniques

NPV

Σ CF□ ÷ (1 + r)^{t} − investment. Accept if above zero. Bharat: +₹10.1L.

IRR

Rate that makes NPV zero. Accept if above cost of capital. Bharat: 17% vs 12%.

Payback

Years to recover investment. Bharat: 3.2 years. Ignores time value and later cash flows.

Profitability index

PV of inflows ÷ investment: 90.1 ÷ 80 = 1.13. Use under capital rationing.

Process: identify opportunity → estimate cash flows → appraise → accept or reject → monitor.

Net Present Value (NPV): Example

NPV = Σ CF□ ÷ (1 + r)^{t} − initial investment. Positive: accept. Zero: indifferent. Negative: reject. NPVs of independent projects add up.

Bharat AutoParts robotic line	Working	Result
Annuity factor (12%, 5 years)	[1 − 1.12^{-5}] ÷ 0.12	3.6048
PV of inflows	₹25,00,000 × 3.6048	₹90,11,938
NPV	₹90,11,938 − ₹80,00,000	+₹10,11,938

Interpretation: the ₹80 lakh investment is expected to create about ₹10.12 lakh of extra value beyond the 12% required return: accept.

Internal Rate of Return (IRR)

IRR is the discount rate at which NPV is zero. Accept if IRR > cost of capital. Found by trial and error, interpolation or the spreadsheet IRR() function.

Bharat AutoParts: rate	PV of inflows	NPV
12%	₹90,11,938	+₹10,11,938
17%	₹80,02,000	≈ +₹2,000
18%	₹78,17,930	−₹1,82,070

Interpolation: 17% + 2,000 ÷ (2,000 + 1,82,070) ≈ 17.01%. IRR of 17% beats the 12% cost of capital: accept, consistent with NPV.

Example 2: When NPV & IRR Disagree

Mutually exclusive, 10% cost of capital	Solar plant	LED retrofit
Investment	₹1,00,00,000	₹10,00,000
Cash flow	₹28,00,000 for 6 years	₹4,50,000 for 4 years
PV of inflows	₹28L × 4.3553 = ₹1,21,94,730	₹4.5L × 3.1699 = ₹14,26,439
NPV	₹21,94,730	₹4,26,439
IRR	17.2%	28.5%

Decision: choose solar. LED has the higher IRR but a small scale; solar adds about five times more rupee value. For mutually exclusive projects let NPV decide. Causes of conflict: scale, reinvestment assumption, multiple IRRs.

Payback & Discounted Payback Period

Year	Cash flow	Cumulative	Discounted at 12%	Cumulative discounted
1	₹25L	₹25L	₹22.32L	₹22.32L
3	₹25L	₹75L	₹17.79L	₹60.05L
4	₹25L	₹100L	₹15.89L	₹75.93L
5	₹25L	–	₹14.19L	₹90.12L

3.2 yrs
Payback: 3 + 5L ÷ 25L

4.29 yrs
Discounted: 4 + 4.07L ÷ 14.19L

Payback is easy to explain but ignores time value and cash flows after recovery.

Example: Bharat AutoParts robotic line

Investment ₹80L. Cash flow ₹25L a year for 5 years. Cost of capital 12%.

₹90.1L
PV of inflows: 25 × 3.6048

+₹10.1L
NPV: accept

≈17%
IRR beats 12%

3.2 yrs
Payback: 80 ÷ 25

1.13
Profitability index: 90.1 ÷ 80

Accept
All four tools agree

Stress test: at a 18% cost of capital, NPV turns negative (PV ≈ ₹78.18L). Ask how sure we are of ₹25L a year.

Summary

Accounting

Assets = Liabilities + Equity. Profit is not cash.

Ratios

Liquidity, leverage, margins, returns. Compare, never read alone.

Costs and CVP

Contribution ÷ fixed cost gives break-even.

TVM

Discount future cash. Rule of 72.

Capital budgeting

NPV first, IRR and PI to support, payback for speed.

Capstone discussion (15 min): a company shows rising profit, falling cash and a current ratio sliding from 1.8 to 1.1. Which statements and ratios would you check, and what would you ask management?

Appendix A: DuPont analysis, ROE in three parts

ROE = Net margin × Asset turnover × Equity multiplier

Asian Paints FY2025	Net margin	Asset turnover	Equity multiplier	ROE
Working	3,588 ÷ 29,553	29,553 ÷ 20,465	20,465 ÷ 18,988	product
Result	12.1%	1.44×	1.08×	18.9%

Two businesses, same ROE	Net margin	Asset turnover	Equity multiplier	ROE
Jewellery retailer (illustration)	20%	0.5×	1.5×	15%
Grocery chain (illustration)	3%	4.0×	1.25×	15%

Interpretation: Asian Paints earns its ROE from margin and turnover, not leverage (multiplier near 1). The illustrations reach the same ROE by margin or by volume. Check which lever drives returns before comparing firms.

Appendix B: profitability index and capital rationing

PI = PV of inflows ÷ investment. When the budget is limited, rank projects by PI and fund down the list. Budget ₹1 crore (figures in ₹ lakh):

Project	Investment	PV of inflows	NPV	PI	Rank
A	40	52	12	1.30	1
B	50	60	10	1.20	2
D	60	69	9	1.15	3
C	30	33	3	1.10	4

Fund A + B: investment ₹90L, NPV ₹22L. A + D would use ₹100L for NPV ₹21L. Bharat AutoParts alone: PI = 90.1 ÷ 80 = 1.13 (above 1 means NPV is positive).